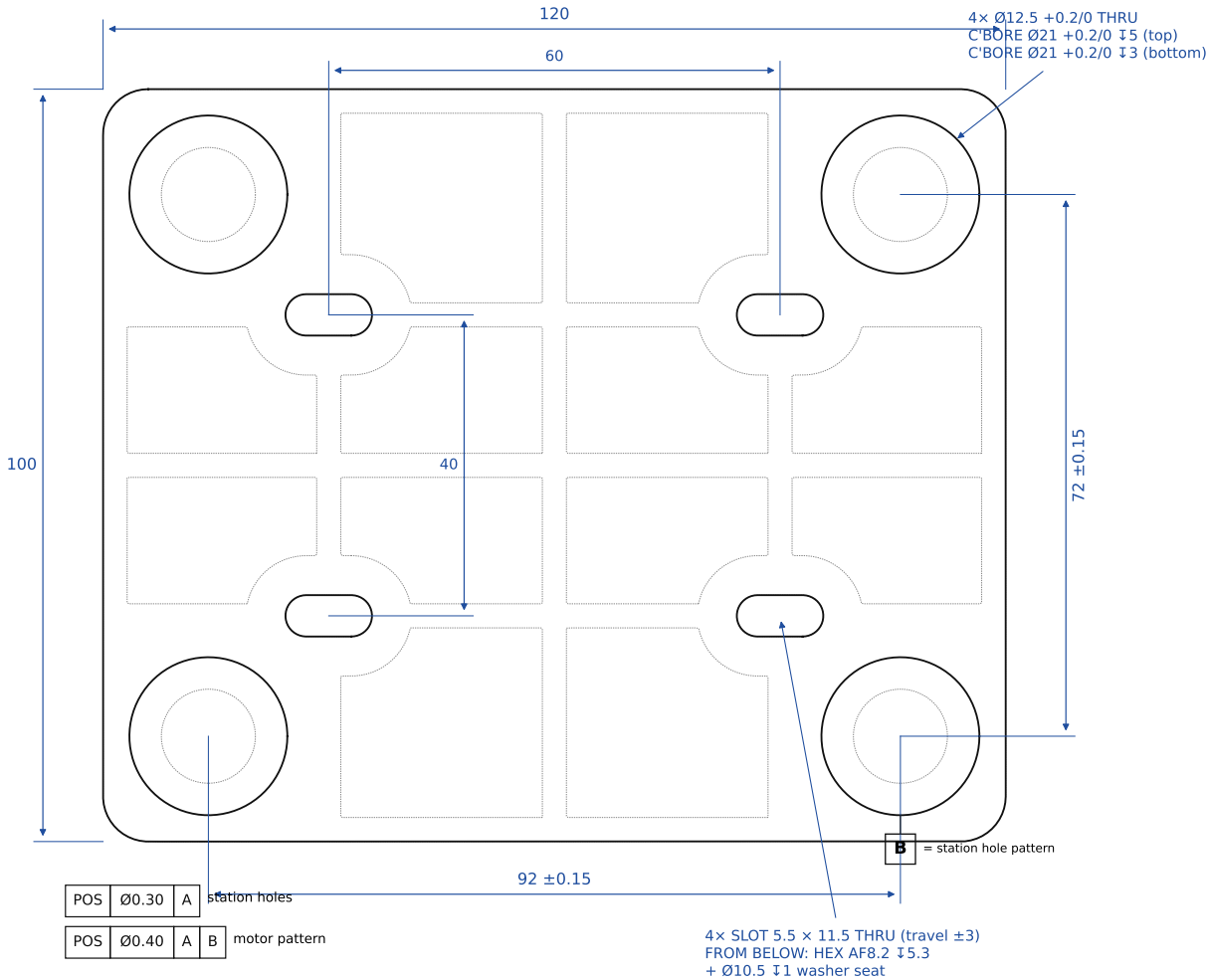
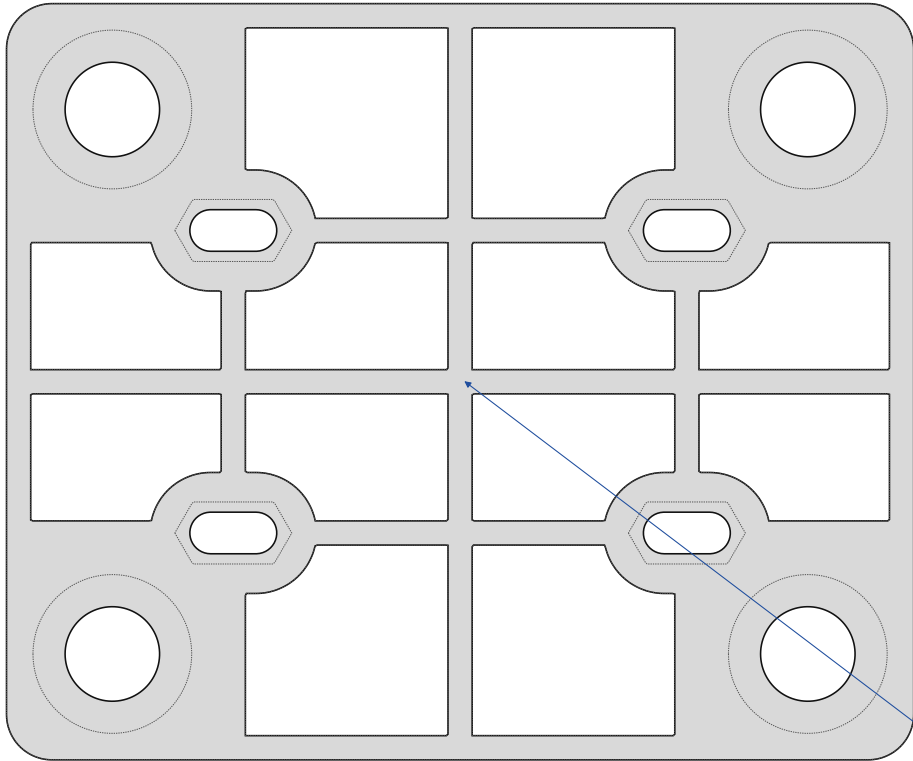


VIEW FROM TOP (MOTOR FACE) 1:1



VIEW FROM BELOW — SECTION Z=7.5 (RIBS) 1:1

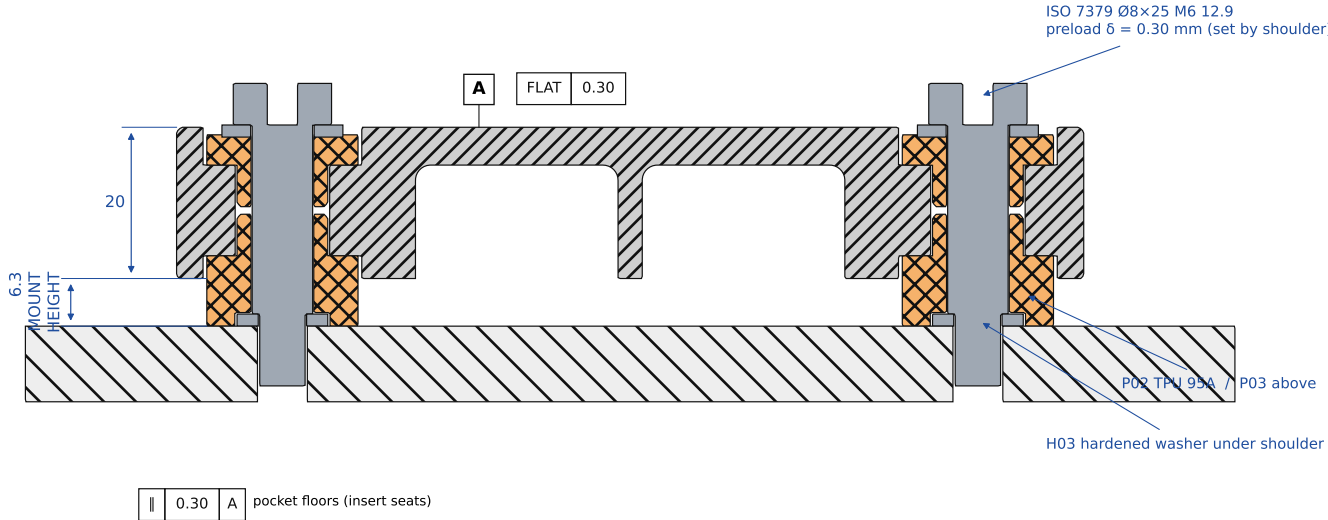


- NOTES
- Units mm. Interpret per ISO 1101 / ISO 128. First-angle projection.
 - General tolerance (FDM): ±0.2 linear, ±1° angular, unless stated.
 - P01: PETG-CF, print deck-down (datum A on bed), 0.2 mm layers, 4 walls, 5 top/bottom, 40 % gyroid. Hardened 0.4 nozzle. Dry filament.
 - P02/P03: TPU 95A, flange on bed, 100 % infill, 0.2 mm, slow (≤ 25 mm/s). Hardness ID: III = notches on flange OD.
 - Ream station Ø12.5 / pocket Ø21 if test-fit is tight. Do NOT enlarge Ø21.
 - Motor M5: prevailing torque + 0.45 N·m. Shoulder screws 6 N·m.
 - Shoulder screws, washers, nuts: METAL ONLY. Never print load-path hardware.
 - Captive design: Ø16 washer > Ø12.5 hole keeps the motor if TPU fails.
 - f_n ≈ 224 Hz (5 kg) / 471 Hz (1 kg). Isolates high-freq noise, NOT shaft-rate (25–50 Hz) vibration. See calculations.
 - Max plate temp 60 °C (PETG-CF). Above: use PA-CF.
 - Prototype part. Validate per tests/ checklist before use.
 - Not for safety-critical or life-safety service.

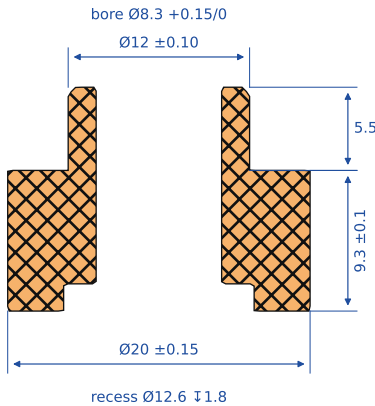
FACTOR OF SAFETY SUMMARY (req. ≥ 2)

TPU lower insert - sustained strain (preload + LC1)	4.3	PASS
TPU lower insert - shock strain (LC3)	9.4	PASS
TPU upper insert - sustained strain (preload)	4.2	PASS
TPU upper insert - rebound strain (LC4)	8.2	PASS
No insert separation in operation (LC2) - down	2.4	PASS
No insert separation in operation (LC2) - up	7.7	PASS
TPU hub radial bearing (1 g lateral)	70.0	PASS
Plate bending, span X = 92 mm (LC3)	2.7	PASS
Plate bending, span Y = 72 mm (LC3)	3.5	PASS
Plate web shear-out at station	51.2	PASS
Motor bolt nut/washer bearing on plate (at spec torque)	2.0	PASS
Motor joint separation (uplift per bolt)	4.5	PASS
Motor joint slip (1 g lateral)	11.1	PASS
Shoulder screw tension (12.9) at spec torque	3.9	PASS
Frame bearing under M6 hardened washer (AI 6061-T6)	3.8	PASS
Shoulder end on hardened washer	3.3	PASS
Thermal mismatch vs hub radial clearance	5.5	PASS
Plate service temperature vs motor case	-	PASS
TPU service temperature vs motor case	-	PASS

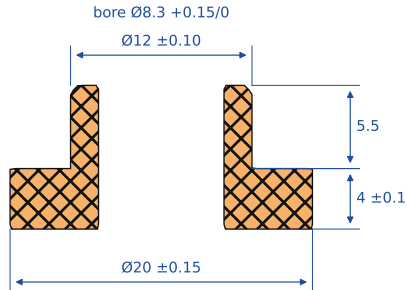
SECTION A-A (THROUGH STATIONS S3/S4) 1:1



P02 LOWER INSERT 2:1
TPU 95A — print as shown



P03 UPPER INSERT 2:1
TPU 95A — print as shown



MASS P01 ≈ 172 g (solid)	REV V1 — INITIAL RELEASE
UNITS mm @ 1st ANGLE	DATE 2026-10-05
MAT PETG-CF / TPU 95A	SCALE 1:1 (DETAIL 2:1)
P/N FLM-MECH-001-V1	SHEET 1/1 A3
PARAMETRIC ANTI-VIBRATION MOTOR MOUNT	
FLEREMIAS ENGINEERING LAB	